DATA 211 – Introduction to Data Science
Topic 2a: Python Basics

Question bank: 40 multiple choice + 10 programming

Questions from DATA 211 / ISE 291 old exams and quizzes (AI_Work). **[new]** marks new questions written in the same style.

Part A: Multiple choice

Q1.

Quiz 1 (T253, Sec 04/05)

What will be the output for the following Python code?

```
str1 = "ISE291SecAB"

print(str1[0:-1])
print(str1[::-1])
```

(A) `ISE291SecAB`
`ISE291SecAB`

(B) `BAceS192ESI`
`BAceS192ESI`

(C) `ISE291SecA`
`IE9ScB`

(D) `ISE291SecA`
`BAceS192ESI`

Answer: (D). `[0:-1]` drops the last character; `[::-1]` reverses the whole string.

My answer

Q2.

Quiz 1 (T253, Sec 04/05)

What is the output of the following Python code?

```
s1 = "American"
s2 = "Japan"

char1 = s1[0]
char2 = s1[int(len(s1) / 2)]
char3 = s1[-1]

char21 = s2[0]
char22 = s2[int(len(s2) / 2)]
char23 = s2[-1]

res = char1 + char21 + char2 + char22 + char3 + char23
print("New String =", res)
```

(A) `New String = AJrpnn`

(B) `New String = AJipnn`

(C) `New String = AinJpn`

(D) Error

Answer: (B). `s1[4]` is `'i'` and `s2[int(2.5)] = s2[2]` is `'p'`, so the pieces are A, J, i, p, n, n.

My answer

Q3.

Quiz 1 (T253, Version F)

What is the output of the following code?

```
M = 'You are amazing.'
M.split('.')
```

(A) `'You are amazing'`

(B) `['You are amazing']`

(C) `['You are amazing', '']`

(D) `You are amazing`

Answer: (C). `split()` returns a list; the text after the final `'.'` is an empty string.

My answer

Q4.

Topic 2 MCQ (T212)

What will be the output of the following Python code snippet?

```
a = ['charA', 'charB', 'charC', 'charD']
b = [c.replace('char','CH ') for c in a]
print(b)
```

(A) `['CH', 'CH', 'CH', 'CH']`

(B) `['CHAR A', 'CHAR B', 'CHAR C', 'CHAR D']`

(C) `['A', 'B', 'C', 'D']`

(D) `['CH A', 'CH B', 'CH C', 'CH D']`

Answer: (D). `replace()` swaps the substring `'char'` for `'CH '` (with a space) in each string.

My answer

Q5.

Topic 2 practice questions

What is the output of the following code?

```
L = list(range(3,12))
L1 = [x//3 for x in L]
print(L1)
```

(A) `[1, 2, 3, 1, 2, 3, 1, 2, 3]`

(B) `[1, 1, 1, 2, 2, 2, 3, 3, 3]`

(C) `{1, 1, 1, 2, 2, 2, 3, 3, 3}`

(D) `[0, 1, 2, 3, 4, 5, 6, 7, 8]`

Answer: (B). `//` is floor division: 3, 4, 5 give 1; 6, 7, 8 give 2; 9, 10, 11 give 3.

My answer

Q6.

Quiz 1 key (Version B)

What is the output?

```
nums = [1, 2, 3]
for i in nums:
    nums.append(i+3)
    if len(nums) > 5:
        break
print(nums)
```

(A) `[1, 2, 3, 4]`

(B) `[1, 2, 3, 4, 6]`

(C) `[1, 2, 3, 4, 5, 6]`

(D) Infinite loop

Answer: (C). The loop appends 4, 5, 6; after appending 6 the length is $6 > 5$, so `break` stops it.

My answer

Q7.

Topic 2 practice questions

What is the output of the following code?

```
Fruits = ['Plum', 'Apple', 'Grapes', 'Banana', 'Watermelon']
Fruits.sort()
i=1
fruits2=[]
for x in Fruits:
   fruits2.append(str(i)+'.'+x)
   i=i+1
print(fruits2)
```

(A) `['1.Apple', '2.Grapes ', '3.Banana ', '4.Watermelon ', '5.Plum ']`

(B) `['1.Apple', '2.Banana', '3.Grapes', '4.Plum', '5.Watermelon']`

(C) `[1.Apple, 2.Banana, 3.Grapes, 4.Plum, 5.Watermelon]`

(D) `["1.Apple", "2.Banana", "3.Grapes", "4.Plum", "5.Watermelon"]`

Answer: (B). The list is sorted alphabetically first, then each name gets its number; Python prints strings with single quotes.

My answer

Q8.

Topic 2 practice questions

Python's `lambda` function is a small anonymous function, which can take:

(A) any number of arguments but can only have one expression

(B) any number of expressions but can only have one value

(C) one argument and can have only one expression

(D) any number of arguments and can have any number of expressions

Answer: (A). A lambda may have many parameters but its body is a single expression.

My answer

Q9.

Topic 2 practice questions

What is the output of the following code?

```
print((lambda x, y, z : x + y**z)(10, 2,3))
```

(A) 18

(B) 36

(C) 16

(D) 19

Answer: (A). $10 + 2^3 = 18$.

My answer

Q10.

Topic 2 practice questions

What is the output of the following code?

```
def f(x): return x**2+3*x-1
Input = [1, 2, 4, 5]
Output= list(map(f, Input))
print(Output)
```

(A) `[4, 10, 27, 39]`

(B) `[3, 9, 28, 40]`

(C) `[3, 9, 27, 39]`

(D) `[1, 3, 9, 27]`

Answer: (C). $f(1) = 3$, $f(2) = 9$, $f(4) = 27$, $f(5) = 39$.

My answer

Q11.

Sample Midterm

The primary purpose of Python's `map()` function is:

(A) To filter elements in a list based on a given condition.

(B) To apply a function to each item in an iterable.

(C) To sort the elements in a list in ascending order.

(D) To concatenate two or more lists together.

Answer: (B). `map(f, iterable)` calls `f` on every element.

My answer

Q12.

Topic 2 practice questions

What is the output of the following code?

```
Input = list(range(6))
Output = list(map((lambda x:x**2 if x%2==0 else 3*x), Input))
print(Output)
```

(A) `[0, 3, 4, 9, 16, 15]`

(B) `[1, 4, 9, 16, 25, 36]`

(C) `[0, 1, 4, 9, 16, 25]`

(D) `{0, 3, 4, 9, 16, 15}`

Answer: (A). Even numbers are squared (0, 4, 16); odd numbers are tripled (3, 9, 15).

My answer

Q13.

Quiz 1 with solution (T252)

Suppose we run:

```
x = ['1','2','3']
y = list(map(int, x))
print(sum(y))
```

What will be printed?

(A) `'123'`

(B) `[1,2,3]`

(C) 6

(D) Error

Answer: (C). `int` converts each string to a number, so the sum is $1 + 2 + 3 = 6$.

My answer

Q14.

Quiz 1 (T253, Sec 03)

Can a single Python list contain elements of different data types?

(A) Yes, lists can contain different data types.

(B) Yes, only if it is created using a specific constructor.

(C) Yes, if we use Python 3 and not Python 2.

(D) No, all elements must be of the same type.

Answer: (A). A list can hold any mix of objects (numbers, strings, other lists, tuples, ...).

My answer

Q15.

Quiz 1 (T253, Sec 03)

What will be the output of the following Python code?

```
my_list = [1, 2, 3]
my_list.append([4, 5])
print(len(my_list))
```

(A) 3

(B) 5

(C) 4

(D) Error

Answer: (C). `append` adds the list `[4, 5]` as one element, giving `[1, 2, 3, [4, 5]]`.

My answer

Q16.

Quiz 1 (T253, Version B)

What is the output of this code?

```
J=[1,2,3,4,5]; K=[5, 6, 7, 8, 9];
len(J+K)
```

(A) 5

(B) 10

(C) 25

(D) None of the above

Answer: (B). + concatenates the lists into one list of 10 elements.

My answer

Q17.

Topic 2 practice questions

What is the output of the following code?

```
x1 = [3, 5, 2, 4, 7, 6, 9, 8]
print(x1[0], x1[3], x1[-1], x1[-3])
```

(A) `3 4 8 6`

(B) `3 2 6 8`

(C) `3 2 8 6`

(D) `0 4 8 6`

Answer: (A). Index 0 is 3, index 3 is 4, `-1` is the last (8), `-3` is the third from the end (6).

My answer

Q18.

Quiz 1 key (Version B)

What is the output?

```
a = [3, 1, 4, 1, 5, 9, 2]
print(a[::-2])
```

(A) `[2, 9, 5, 1]`

(B) `[3, 1, 4, 1]`

(C) `[9, 5, 1]`

(D) `[2, 5, 4, 3]`

Answer: (D). Step -2 starts at the last element and takes every second element going backwards: 2, 5, 4, 3.

My answer

Q19. Quiz 1 Sec 4 (Topics 2 & 3 file)

How would you sort a list of fruits in descending order?

(A) `fruits.sort(reverse=True)`

(B) `sorted(fruits)`

(C) `fruits.reverse()`

(D) `sorted(fruits, reverse=False)`

Answer: (A). `reverse=True` sorts from largest to smallest; `reverse()` only flips the current order.

My answer

Q20. Midterm practice questions 2

Which of the following is TRUE about list comprehensions?

(A) They always add an index number with each member of a list.

(B) They provide a concise way to define a for loop to create a list.

(C) They must contain an if statement.

(D) They can be used only with numeric data types.

Answer: (B). A comprehension is a compact for loop that builds a list; the `if` part is optional.

My answer

Q21. Topic 2 practice questions

What is the output of the following code?

```
L=list(range(0,6))
L1=[x**2 for x in L if x%2==0]
L2=[x**2 for x in L if x%2==1]
print(L1)
print(L2)
```

(A) `[0, 4, 16]` and `[1, 9, 25]`

(B) `{0, 4, 16}` and `{1, 9, 25}`

(C) `[0, 2, 4]` and `[1, 3, 5]`

(D) `[0, 4, 16, 36]` and `[1, 9, 25]`

Answer: (A). `range(0,6)` stops at 5: even squares 0, 4, 16 and odd squares 1, 9, 25.

My answer

Q22. Midterm practice questions 2

Which of the following lines of code will generate a list of all numbers from 1 to 20 (both inclusive) which are NOT divisible by 3?

(A) `[x for x in range(20) if x%3 != 0]`

(B) `[x for x in range(1, 21) if x%3 != 0]`

(C) `[x for x in range(1,20,3)]`

(D) `[x for x in range(0,21) if x%3 == 0]`

Answer: (B). `range(1, 21)` covers 1 to 20, and `x%3 != 0` keeps the numbers not divisible by 3.

My answer

Q23. Midterm practice questions 2

What is the output of the following code?

```
A = [2 * x if x % 2 == 1 else x for x in range(5,1,-1)]
B = list(map(lambda x : x - 2, A))
print(B)
```

(A) `[0, 0, 4, 2]`

(B) `[3, 6, 1, 2]`

(C) `[12, 6, 8, 4]`

(D) `[8, 2, 4, 0]`

Answer: (D). `range(5,1,-1)` is 5, 4, 3, 2, so `A = [10, 4, 6, 2]`; subtracting 2 gives `[8, 2, 4, 0]`.

My answer

Q24. Quiz 1 key (T251)

What output will you get with this code?

```
List1 = [x**3 if x%2==0 else x for x in range(1,11)]
print(List1)
```

(A) `[1, 8, 3, 64, 5, 216, 7, 512, 9, 1000]`

(B) `[1, 6, 3, 12, 5, 18, 7, 24, 9, 30]`

(C) `[1, 2, 27, 4, 125, 6, 343, 8, 729, 10]`

(D) `[1, 8, 3, 64, 5, 216, 7, 512, 9, 1000, 11]`

Answer: (A). Even numbers 2 to 10 are cubed, odd numbers stay the same, and `range(1,11)` stops at 10.

My answer

Q25.

Quiz 1 (T253, Sec 04/05)

What is the output of the following Python code?

```
NmList = ['Jawad', 'Yaseen', 'Aymen', 'Mohammad', 'Jamal', 'Ajmal', 'Basem', 'Walid', 'Emad'
    , 'Fayed', 'Aman', 'Rashad']
newList = [x for x in NmList if x[-1]=='d']
print(newList)
```

(A) ['Jawad', 'Mohammad', 'Walid', 'Emad', 'Fayed', 'Rashad']

(B) {'Jawad', 'Mohammad', 'Walid', 'Emad', 'Fayed', 'Rashad'}

(C) (['Jawad', 'Mohammad', 'Walid', 'Emad', 'Fayed', 'Rashad'])

(D) Error

Answer: (A). `x[-1]` is the last letter; the comprehension keeps the names ending in `'d'` and returns a list.

My answer

Q26.

Quiz 1 sample (Topics 2 & 3)

Given the list

```
fruits = [['Apple', 'Banana', 'Orange'],
          ['Watermelon', 'Plum', 'Grapes', 'Kiwi'],
          ['Strawberry', 'Pear', 'Mango']]
```

which of the following codes will generate the list
`['Apple', 'Banana', 'Orange', 'Strawberry', 'Pear', 'Mango']`?

(A) `allFruits = [y for innerList in fruits for y in innerList if len(y)<4]`

(B) `allFruits = [y for innerList in fruits for y in innerList if len(innerList)>3]`

(C) `allFruits = [y for innerList in fruits for y in innerList if len(innerList)<4]`

(D) `allFruits = [y for innerList in fruits for y in innerList if len(y)>3]`

Answer: (C). Only the inner lists with fewer than 4 items (the first and third) are kept; their fruits are flattened in order.

My answer

Q27.

Sample Midterm

Which of the following is TRUE about `enumerate`?

(A) For the enumeration, the start indexing value by default is 1.

(B) Enumerate is used to combine elements of any two lists.

(C) Enumerate is used to iterate over a sequence while keeping track of the index.

(D) Enumerate is used to reverse the order of elements in a sequence.

Answer: (C). `enumerate` yields `(index, value)` pairs; the default start is 0.

My answer

Q28.

Topic 2 MCQ (T212)

What will be the output of the following Python code snippet?

```
a = ['A', 'B', 'C', 'D']
b = [v + str(i) for i,v in enumerate(a)]
print(b)
```

(A) `['A0', 'B1', 'C2', 'D3']`

(B) `('A0', 'B1', 'C2', 'D3')`

(C) `['0A', '1B', '2C', '3D']`

(D) `('0A', '1B', '2C', '3D')`

Answer: (A). `i` is the index and `v` the value; `v + str(i)` puts the letter first, and a list comprehension gives a list.

My answer

Q29.

Quiz 1 (T253, Version B)

What is the output of the following code?

```
A=['a', 'b','c']; B=[4, 5, 6];
Z=list(zip(A,enumerate(B,start=10)))
print(Z)
```

(A) `[('a', (10, 4)), ('b', (11, 5)), ('c', (12, 6))]`

(B) `[('a',10, 4), ('b',11, 5), ('c',12, 6)]`

(C) `[('a', (0, 4)), ('b', (1, 5)), ('c', (2, 6))]`

(D) `[(10, ('a', 4)), (11, ('b', 5)), (12, ('c', 6))]`

Answer: (A). `enumerate(B, start=10)` gives the tuples `(10,4)`, `(11,5)`, `(12,6)`, and `zip` pairs each one with a letter.

My answer

Q30.

Topic 2 MCQ (T212)

What will be the output of the following Python code snippet?

```
a =(1,2,3)
b =('A','B','C')
c = zip(a,b)
print(list(c))
```

(A) `((1,2,3),('A','B','C'))`

(B) `[(1,2,3),('A','B','C')]`

(C) `[(1, 'A'), (2, 'B'), (3, 'C')]`

(D) `([1, 'A'], [2, 'B'], [3, 'C'])`

Answer: (C). `zip` pairs items in the same position into tuples; `list()` collects them.

My answer

Q31.

Topic 2 practice questions

What is the output of the following code?

```
Fruits = ['Apple', 'Kiwi', 'Banana', 'Watermelon', 'Dates']
Fruits2 = sorted(Fruits, reverse=True)
length = [len(v) for v in Fruits2]
ZipZap = list(zip(Fruits2,length))
print(ZipZap)
```

(A) `[(Watermelon, 10), (Kiwi, 4), (Dates, 5), (Banana, 6), (Apple, 5)]`

(B) `[('Apple', 5), ('Banana', 6), ('Dates', 5), ('Kiwi', 4), ('Watermelon', 10)]`

(C) `[('Watermelon'=10), ('Kiwi'=4), ('Dates'=5), ('Banana'=6), ('Apple'=5)]`

(D) `[('Watermelon', 10), ('Kiwi', 4), ('Dates', 5), ('Banana', 6), ('Apple', 5)]`

Answer: (D). The names are sorted Z to A and each one is paired with its length as a tuple.

My answer

Q32.

Quiz 1 (T253, Sec 03)

Which of the following defines a Dictionary?

(A) `{"name": "apple", "color": "green"}`

(B) `{"apple", "banana", "cherry"}`

(C) `("apple", "banana", "cherry")`

(D) `["apple", "banana", "cherry"]`

Answer: (A). A dictionary holds `key: value` pairs in curly braces; (B) is a set, (C) a tuple, (D) a list.

My answer

Q33. Sample Midterm

In Python, which of the following statements is FALSE about keys in the dictionary?

(A) A given key can appear in a dictionary twice.

(B) Re-assigning a new value to an existing key will override the first/previous value.

(C) Dictionary keys can be of any type, but they must be immutable.

(D) Duplicate keys are not allowed.

Answer: (A). Keys are unique; assigning to an existing key replaces its value.

My answer

Q34. Topic 2 MCQ (T212)

Which of the following statements about dictionary values is false?

(A) More than one key can have the same value

(B) The values of the dictionary can be accessed as `dic_name[key]`

(C) Values of a dictionary must be unique

(D) Values of a dictionary can be of any type

Answer: (C). Only the keys must be unique; values can repeat and be of any type.

My answer

Q35. Topic 2 MCQ (T212)

What will be the output of the following Python code snippet?

```
d = {}
d['a'] = 1
d['b'] = [2,3,4]
d['a'] = 2
print(d)
```

(A) `{'a': [2, 3, 4], 'b': 2}`

(B) `{'a': [2, 3, 4], 'b': 1}`

(C) `{'a': 1, 'b': [2, 3, 4]}`

(D) `{'a': 2, 'b': [2, 3, 4]}`

Answer: (D). Assigning to a new key adds it; assigning `d['a'] = 2` again replaces the old value 1.

My answer

Q36.

Sample Midterm

What will be the output of the following Python code?

```
d = {'length':29, 'width':15, 'height':74}
print(list(d.keys()))
```

(A) `('length', 'width', 'height')`

(B) `['length', 'width', 'height']`

(C) `[29, 15, 74]`

(D) `['length':29, 'width':15, 'height':74]`

Answer: (B). `keys()` gives the keys, and `list()` turns them into a list.

My answer

Q37.

Quiz 1 (T253, Sec 03)

What will be the output for the following Python code?

```
nums = [2, 4, 6, 8]
result = dict(zip(nums, map(lambda x: x * 3, nums)))
print(result)
```

(A) `{2:3, 4:6, 6:9, 8:12}`

(B) `{2: 6, 4: 12, 6: 18, 8: 24}`

(C) `[6, 12, 18, 24]`

(D) `[2, 6, 4, 12, 6, 18, 8, 24]`

Answer: (B). Each number is paired with three times itself, and `dict()` turns the pairs into key: value entries.

My answer

Q38.

Topic 2 practice questions

What is the output of the following code?

```
Fruits = ['Apple', 'Kiwi', 'Banana', 'Watermelon', 'Dates']
Prices=[10, 12, 6, 4, 60]
Fruits2 = sorted(Fruits, reverse=True)
Book = dict(zip(Fruits2,Prices))
print(Book)
```

(A) `{'Apple': 10, 'Banana': 12, 'Dates': 6, 'Kiwi': 4, 'Watermelon': 60}`

(B) `['Apple': 10, 'Banana': 12, 'Dates': 6, 'Kiwi': 4, 'Watermelon': 60]`

(C) `{'Watermelon': 10, 'Kiwi': 12, 'Dates': 6, 'Banana': 4, 'Apple': 60}`

(D) `('Watermelon': 10, 'Kiwi': 12, 'Dates': 6, 'Banana': 4, 'Apple': 60)`

Answer: (C). The names are sorted Z to A and then paired in order with the unsorted prices. (In the original, option C used square brackets, so no option was correct. Option C now uses braces.)

My answer

Q39.

Quiz 1 with solution (T252)

What is the result of:

```
numbers = [1,2,3]
squared = {n: n**2 for n in numbers if n%2==1}
print(squared)
```

(A) `{1:1, 2:4, 3:9}`

(B) `{1:1, 3:9}`

(C) `[1,9]`

(D) Error

Answer: (B). This dictionary comprehension keeps only the odd numbers 1 and 3 as keys, with their squares as values.

My answer

Q40.

Sample Midterm

What will be the output of the following Python code?

```
numbers = list(range(10))
SQR = list(map((lambda x:x**2 if x%2==0 else None), numbers))
print(SQR)
```

(A) `[0, None, 4, None, 16, None, 36, None, 64, None, 100]`

(B) `[0, None, 4, None, 16, None, 36, None, 64, None]`

(C) `[0:0, 1:None, 2:4, 3:None, 4:16, 5:None, 6:36, 7:None, 8:64, 9:None]`

(D) `[0:0, 1:None, 2:4, 3:None, 4:16, 5:None, 6:36, 7:None, 8:64, 9:None, 10:100]`

Answer: (B). `range(10)` stops at 9; even numbers are squared and odd numbers give `None`. (The original wrote `none`, which would raise a NameError; this version uses `None`.)

My answer

Part B: Programming

Q41.

Midterm Set-1

Create a list of all powers of 3 from 0 to 8, using list comprehension. Do NOT use the NumPy library. The output should look like this: `[1, 3, 9, 27, ...]`

Answer:

```
powers = [3**x for x in range(9)]
print(powers)
# [1, 3, 9, 27, 81, 243, 729, 2187, 6561]
```

My answer

Q42.

Midterm 213

Create a list of numbers, say `list1`, in the range 100. Create another new list from the numbers in `list1`, say `list2`, that contains only the numbers that are divisible by *both* 3 and 10. You must use list comprehension. Do NOT use the NumPy library. (Output hint: `[0, 30, 60, 90]`)

Answer:

```
list1 = [i for i in range(100)]
list2 = [x for x in list1 if x % 3 == 0 and x % 10 == 0]
print(list2)
# [0, 30, 60, 90]
```

My answer

Q43.

Midterm Set-1

Write a program that does the following:

1. asks the user to input a number between 10 and 100 (both inclusive),
2. checks if the input is valid (it is between 10 and 100),
3. if the input is NOT valid, prints `"out of range"` and exits,
4. if the input is valid, prints all the factors of the input number and exits.

For example, if you input 96 it will print `[1, 2, 3, 4, 6, 8, 12, 16, 24, 32, 48, 96]`. (Note: if X is a factor of Y, then Y is completely divisible by X.)

Answer:

```
n = int(input("Enter a number between 10 and 100: "))
if 10 <= n <= 100:
    factors = [i for i in range(1, n + 1) if n % i == 0]
    print(factors)
else:
    print("out of range")
# input 96 -> [1, 2, 3, 4, 6, 8, 12, 16, 24, 32, 48, 96]
# input 5  -> out of range
```

My answer

Q44.

Midterm Set-1

Consider the following string:

```
ids = 's202731518 s202113599 s2027@66875 s202689361 s202727303 s202487613 s20254\t2892
    s202514475 s202573900 s202383642'
```

Do the following steps:

1. Obtain a string from the above string by removing all special characters (like `'@'` and `'\t'`). Do NOT remove the white spaces.
2. Create a list of the 9-digit student IDs from that string (e.g. `'s202731518'` is a student ID).
3. For each element in the list, add `"@kfupm.edu.sa"` at its end (e.g. `'s202731518@kfupm.edu.sa'`).
4. Merge all the email addresses into a single string separated by semicolons, in the format `"s202731518@kfupm.edu.sa;s202113599@kfupm.edu.sa;...;s202383642@kfupm.edu.sa"`.

Answer:

```
ids = 's202731518 s202113599 s2027@66875 s202689361 s202727303 s202487613 s20254\t2892
    s202514475 s202573900 s202383642'
clean = ids.replace('\t', '').replace('@', '')   # step 1
id_list = clean.split(' ')                       # step 2
emails = [s + '@kfupm.edu.sa' for s in id_list]  # step 3
```

```
output = ';'.join(emails)                    # step 4
print(output)
```

My answer

Q45.

Quiz #01 (notebook)

Do not use the NumPy library for this question. Convert the following text into a list of words:

```
text = "today we are attending ISE 291 quiz"
```

Output hint: `['today', 'we', 'are', 'attending', 'ISE', '291', 'quiz']`

Answer:

```
text = "today we are attending ISE 291 quiz"
words = text.split(' ')
print(words)
# ['today', 'we', 'are', 'attending', 'ISE', '291', 'quiz']
```

My answer

Q46.

Midterm Set-1

Consider the function $F(x) = 5x^2 - 3x + 2$. Create a dictionary using the following steps:

1. Compute $F(x)$ for the x-values `[12, 5, 12, 2, 8, 1, 13, 8, 9, 2]`.
2. Create a dictionary for which x is the key and $F(x)$ is the value.

The output should look like: {`12: 686, 5: 112, ...`}

Answer:

```
def F(x):
    return 5*x**2 - 3*x + 2

nums = [12, 5, 12, 2, 8, 1, 13, 8, 9, 2]
output = dict(zip(nums, map(F, nums)))
print(output)
# {12: 686, 5: 112, 2: 16, 8: 298, 1: 4, 13: 808, 9: 380}
# (repeated x-values become one key, because keys are unique)
```

My answer

Q47. Quiz 1 Part 2 (notebook)

Given the following lists of student names and their scores in two quizzes:

```
names = ["Saleh", "Munir", "Ahmed", "Saad", "Fahd", "Fayed", "Amr",
         "Ali", "Naser", "Nizar", "Amjd", "Abdo", "Badr", "Syed"]
quiz_1 = [8.5, 6.4, 7.2, 7.3, 6.5, 6.0, 6.1, 7.2, 6.6, 8.0, 8.9, 7.7, 5.0, 6.3]
quiz_2 = [6.3, 8.4, 6.2, 7.4, 6.5, 7.0, 6.1, 7.2, 7.7, 8.1, 7.9, 8.7, 7.0, 5.3]
```

Using `map` with a `lambda` function, compute the average of the two quizzes for each student. Then create and print a dictionary whose keys are the student names and whose values are the averages.

Answer:

```
average = list(map(lambda x, y: (x + y) / 2, quiz_1, quiz_2))
names_dict = dict(zip(names, average))
print(names_dict)
# {'Saleh': 7.4, 'Munir': 7.4, 'Ahmed': 6.7, 'Saad': 7.35, ..., 'Syed': 5.8}
```

My answer

Q48.

HW 1 (T253)

Using a list comprehension, create a list `L1` of the integers from 1 to 78. Sort `L1` in descending order. Then, using `enumerate(L1)` (index `j`, value `k`) in a list comprehension, create a list `L2` that keeps only the even values `k` and replaces each one with `k - 1` if `j > k`, and with `k + 1` otherwise. Print `L1` and `L2`.

Answer:

```
L1 = [x for x in range(1, 79)]
L1.sort(reverse=True)
L2 = [k - 1 if j > k else k + 1 for j, k in enumerate(L1) if k % 2 == 0]
print(L1)
print(L2)
# L1: [78, 77, 76, ..., 2, 1]
# L2: [79, 77, ..., 43, 41, 37, 35, ..., 3, 1]
```

My answer

Q49.

HW 1 (T253)

Two texts are given:

```
text1 = "cos30 cos45 cos60 cos90 sin30 sin45 sin60 sin90 tan30 tan45"
text2 = "0.866025 0.707107 0.5 0 0.5 0.707107 0.866025 1 0.57735 1"
```

Split each text into a list of words. Use `zip()` and `dict()` to build `My_dictionary`, which pairs each trigonometric function with its value, and print it. Then print the total number of words in `text1` and

the total number of keys in `My_dictionary`, and compare the two counts to check whether any key was duplicated.

Answer:

```
funcs = text1.split()
values = text2.split()
My_dictionary = dict(zip(funcs, values))
print(My_dictionary)
print("The total number of words in text1 :", len(funcs))
print("The total number of keys in My_dictionary :", len(My_dictionary.keys()))
if len(funcs) == len(My_dictionary.keys()):
    print("No duplicate keys")
else:
    print("Some keys were duplicated")
# both counts are 10 -> no duplicate keys
```

My answer

Q50.

HW 1 (T253)

A restaurant sells the following items:

```
foods = ['Shawarma', 'Chicken', 'Meat', 'Kabsa']
costs = [14, 18, 22, 25]
```

Use `zip()` and `dict()` to pair each food item with its cost. Then loop over `dict.items()` and print one line per item in the form `The price of Shawarma is 14.`

Answer:

```
menu = dict(zip(foods, costs))
for item, price in menu.items():
    print(f"The price of {item} is {price}.")
# The price of Shawarma is 14.
# The price of Chicken is 18.
# The price of Meat is 22.
# The price of Kabsa is 25.
```

My answer